

P. N. KARTHIK

[EMAIL](#)[WEBSITE](#)[LINKEDIN](#)[GOOGLE SCHOLAR](#)

RESEARCH INTERESTS

multi-armed bandits, federated learning, differential privacy, reinforcement learning, information theory, statistics

EXPERIENCE

- FEB'26 - PRESENT **ASSISTANT PROFESSOR GRADE-I**
Department of Artificial Intelligence, IIT Hyderabad
- FEB'24 - JAN'26 **ASSISTANT PROFESSOR GRADE-II**
Department of Artificial Intelligence, IIT Hyderabad
DATE OF JOINING: 26 February 2024
- JAN'22 - JAN'24 **RESEARCH FELLOW**
Institute of Data Science, National University of Singapore
SUPERVISOR: Prof. [Vincent Y. F. Tan](#)
- MAY'23 - JUN'23 **VISITING RESEARCHER**
Technion Israel Institute of Technology, Haifa, Israel
HOST: Prof. [Nir Weinberger](#)
- NOV'19 - MAR'20 **RESEARCH INTERN**
Netradyne Technology India Pvt. Ltd., Bengaluru
In collaboration with Bengaluru Metropolitan Transport Corporation
ACADEMIC MENTOR: Prof. [Rajesh Sundaresan](#)
INDUSTRY MENTOR: [Pratik Verma](#)
- AUG'19 - DEC'19 **GRADUATE TEACHING ASSISTANT**
AUG'18 - DEC'18 *Department of ECE, Indian Institute of Science, Bengaluru*
AUG'17 - DEC'17 COURSES: Information Theory (E2 201), Random Processes (E2 202)
INSTRUCTORS: Prof. [Himanshu Tyagi](#), Prof. [Utpal Mukherji](#), Prof. [Parimal Parag](#)
- AUG'14 - JUN'15 **PROJECT ASSISTANT**
Department of ECE, Indian Institute of Science, Bengaluru
SUPERVISOR: Prof. [Chandra R. Murthy](#)

EDUCATION

- JUL'15 - MAR'22 **DOCTOR OF PHILOSOPHY AND MASTER OF SCIENCE (ENGINEERING),**
[Indian Institute of Science](#), Bengaluru
DEPARTMENT: Electrical Communication Engineering
THESIS: Sequential Controlled Sensing to Detect an Anomalous Process
SUPERVISOR: Prof. [Rajesh Sundaresan](#)
GPA: 7.0/8.0
- AUG'10 - JUL'14 **BACHELOR OF ENGINEERING,**
[R V College of Engineering](#), Bengaluru
MAJOR: Electronics and Communications
GPA: 9.72/10.00 (**RANK 2** among 140 students)

PUBLICATIONS

PREPRINTS

1. *Almost Asymptotically Optimal Active Clustering Through Pairwise Observations* [arxiv](#)
Rachel S. Y. Teo, **P. N. Karthik**, Ramya Korlakai Vinayak, and Vincent Y. F. Tan
Under review at IEEE Transactions on Information Theory
2. *Learning to Detect an Odd Restless Markov Arm with a Trembling Hand* [arxiv](#)
P. N. Karthik and Rajesh Sundaresan
3. *Axiomatic Characterisation of Projection Rules: An Open Question* [draft](#)
P. N. Karthik and Rajesh Sundaresan

JOURNAL PUBLICATIONS

1. *Optimal Best Arm Identification with Fixed Confidence in Restless Bandits* [arxiv](#)
P. N. Karthik, Vincent Y. F. Tan, Arpan Mukherjee, and Ali Tajer
Accepted, [IEEE Transactions on Information Theory](#), 2024
Doi: [10.1109/TIT.2024.3419924](#)
2. *Federated Best Arm Identification with Heterogeneous Clients*
Chen Zhirui, **P. N. Karthik**, Vincent Y. F. Tan, and Yeow Meng Chee
[IEEE Transactions on Information Theory](#), volume 70, number 6, pp. 4258-4279, 2024
Doi: [10.1109/TIT.2023.3338027](#)
3. *Bus Priority lane in Bengaluru: A Study of its Effectiveness and Driver Stress*
P. N. Karthik, Nihesh Rathod, Sarath Yasodharan, Wilson Lobo, Ajeesh Sahadevan, Rajesh Sundaresan and Pratik Verma
Special Issue on Sustainable City Transportation in the Indian Subcontinent, [Transport Policy](#), 2023.
Doi: [10.1016/j.tranpol.2023.04.018](#)
4. *Best Arm Identification in Restless Markov Multi-Armed Bandits*
P. N. Karthik, Kota Srinivas Reddy, and Vincent Y. F. Tan
[IEEE Transactions on Information Theory](#), volume 69, number 5, pp. 3240-3262, 2023.
Doi: [10.1109/TIT.2022.3230939](#)
5. *Detecting an Odd Restless Markov Arm with a Trembling Hand*
P. N. Karthik and Rajesh Sundaresan
[IEEE Transactions on Information Theory](#), volume 67, number 8, pp. 5230-5258, 2021.
Doi: [10.1109/TIT.2021.3075021](#)
6. *Learning to Detect an Odd Markov Arm*
P. N. Karthik and Rajesh Sundaresan
[IEEE Transactions on Information Theory](#), volume 66, number 7, pp. 4324-4348, 2020.
Doi: [10.1109/TIT.2020.2972875](#)

DOCTORAL DISSERTATION

- *Sequential Controlled Sensing to Detect an Anomalous Process* [pdf](#) [source](#) [etd iisc](#)
Ph.D. thesis, Department of ECE, Indian Institute of Science, NOV 2021

CONFERENCE PUBLICATIONS

1. *Link Adaptation Using Joint-Thompson Sampling*
Vignatha Vinjam, Manjunath Kolavennu, Myna Vajha, and **P. N. Karthik**
International Conference on Signal Processing and Communications ([SPCOM 2026](#)), JUL 2026
2. *Asymptotically Optimal Sequential Testing with Markovian Data*
Alhad Sethi, Kavali Sofia Sagar, Shubhada Agrawal, Debabrota Basu, and **P. N. Karthik**
International Conference on Machine Learning ([ICML 2026](#)), JUL 2026
3. *Almost Asymptotically Optimal Active Clustering Through Pairwise Observations*
Rachel S. Y. Teo, **P. N. Karthik**, Ramya Korlakai Vinayak, and Vincent Y. F. Tan
IEEE International Symposium on Information Theory ([ISIT 2026](#)), JUN 2026
4. *Optimal Multi-Objective Best Arm Identification with Fixed Confidence*
Chen Zhirui, **P. N. Karthik**, Yeow Meng Chee, and Vincent Y. F. Tan
International Conference on Artificial Intelligence and Statistics ([AISTATS 2025](#)), MAY 2025

5. *Best Arm Identification with Arm Erasures*
Kota Srinivas Reddy, **P. N. Karthik**, and Vincent Y. F. Tan
IEEE International Symposium on Information Theory (ISIT 2024), JUL 2024
Doi: [10.1109/ISIT57864.2024.10619688](https://doi.org/10.1109/ISIT57864.2024.10619688)
6. *Fixed-Budget Differentially Private Best Arm Identification*
Chen Zhirui, **P. N. Karthik**, Yeow Meng Chee, and Vincent Y. F. Tan
International Conference on Learning Representations (ICLR 2024), MAY 2024
7. *Best Arm Identification in Bandits with Limited Precision Sampling*
Kota Srinivas Reddy, **P. N. Karthik**, Nikhil Karamchandani, and Jayakrishnan Nair
IEEE International Symposium on Information Theory (ISIT 2023), JUN 2023
Doi: [10.1109/ISIT54713.2023.10206610](https://doi.org/10.1109/ISIT54713.2023.10206610)
8. *Almost Cost-Free Communication in Federated Best Arm Identification*
Kota Srinivas Reddy, **P. N. Karthik**, and Vincent Y. F. Tan
Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI 2023), Washington D.C., FEB 2023.
Doi: [10.1609/aaai.v37i7.26010](https://doi.org/10.1609/aaai.v37i7.26010)
9. *Best Restless Markov Arm Identification*
P. N. Karthik, Kota Srinivas Reddy, and Vincent Y. F. Tan
Proceedings of the 2022 IEEE Information Theory Workshop (ITW 2022), IIT Mumbai, Nov 2022.
Doi: [10.1109/ITW54588.2022.9965908](https://doi.org/10.1109/ITW54588.2022.9965908)
10. *Learning to Detect an Odd Restless Markov Arm*
P. N. Karthik and Rajesh Sundaresan
Proceedings of the 2021 IEEE International Symposium on Information Theory (ISIT 2021), virtual, JUL 2021.
Doi: [10.1109/ISIT45174.2021.9518083](https://doi.org/10.1109/ISIT45174.2021.9518083)
11. *Detecting an Odd Restless Markov Arm with a Trembling Hand*
P. N. Karthik and Rajesh Sundaresan
Proceedings of the 2020 IEEE International Symposium on Information Theory (ISIT 2020), virtual, JUN 2020.
Doi: [10.1109/ISIT44484.2020.9174397](https://doi.org/10.1109/ISIT44484.2020.9174397)
12. *Learning to Detect an Odd Markov Arm*
P. N. Karthik and Rajesh Sundaresan
Proceedings of the 2019 IEEE International Symposium on Information Theory (ISIT 2019), Paris, France, JUL 2019.
Doi: [10.1109/ISIT.2019.8849807](https://doi.org/10.1109/ISIT.2019.8849807)
13. *On The Equivalence of Projections in Relative α -Entropy and Rényi Divergence*
P. N. Karthik and Rajesh Sundaresan
Proceedings of the 34th National Conference on Communications (NCC 2018), IIT Hyderabad, FEB 2018.
Doi: [10.1109/NCC.2018.8599980](https://doi.org/10.1109/NCC.2018.8599980)
14. *Model-Based Interference Cartography and Visualization*
P. N. Karthik, Raksha Ramakrishna, Geethu Joseph, Chandra R. Murthy, Joyson Sebastian, and Neelesh B. Mehta
Proceedings of the 22nd National Conference on Communications (NCC 2016), IIT Guwahati, MAR 2016.
Doi: [10.1109/NCC.2016.7561174](https://doi.org/10.1109/NCC.2016.7561174)

TEACHING AT IIT HYDERABAD

- *AI5090: Stochastic Processes: Theory and Applications* (JAN–APR 2026)
Number of students enrolled: 23
Instructor feedback: 4.81/5
- *AI1013: Programming for AI* (FEB–APR 2026)
Number of students enrolled: 40
Instructor feedback: 4.56/5
- *AI5030: Probability and Stochastic Processes* (JUL–NOV 2025)
Number of students enrolled: 94
Instructor feedback: 4.77/5
- *AI1013: Programming for AI* (FEB–APR 2025)
Number of students enrolled: 36
Instructor feedback: 4.74/5

- *AI5090: Stochastic Processes: Theory and Applications* (JAN–APR 2025)
Number of students enrolled: 33
Instructor feedback: 4.95/5
- *CS6660: Mathematical Foundations of Data Science* (JUL–NOV 2024)
Number of students enrolled: 25
Instructor feedback: 4.8/5
- *AI5030: Probability and Stochastic Processes* (JUL–NOV 2024)
Number of students enrolled: 87
Instructor feedback: 4.84/5
- *AI1104: Programming for AI* (MAR–APR 2024)
Number of students enrolled: 73
Instructor feedback: 4.19/5

AWARDS AND HONORS

- ANRF Prime Minister Early Career Research Grant (PMECRG), 2026-2029
- Faculty Teaching Excellence Award 2025, presented by IIT Hyderabad
- First place in the 100 seconds competition organised by INAE Kanpur Chapter
- Best paper award at the 2020 EECS Research Students' Symposium, Indian Institute of Science
- Best 3-minute presentation, ECE Students' Seminar Series, Department of ECE, Indian Institute of Science
- Rank 136 (among the top 0.01%) in the 2015 Graduate Aptitude Test in Engineering
- Infineon India scholarship for securing rank 2 in the 2011 Visvesvaraya Technological University examinations.
- Rank 23 (among the top 0.02%) in the 2010 Karnataka Common Entrance Test

SERVICE AT IIT HYDERABAD

- Member, Space Committee (SEP 2025 - PRESENT)
- Faculty Advisor (FA) for BTech 2025 batch of students (JUN 2025 - PRESENT)
- Faculty Advisor (FA) for BTech 2024 batch of students (JUN 2024 - PRESENT)
- Member, Library Committee, Department of AI (MAR 2025 - PRESENT)
- Member, Faculty Search Committee (FSC), Department of AI (JAN 2025 – PRESENT)
- Faculty Advisor (FA) for MTech 2024 batch of students (AUG 2024 – PRESENT)
- Faculty-in-Charge, Graduate Admissions, Department of AI (OCT 2024 – PRESENT)
- Faculty Coordinator, [Advanced Executive Program in AI and Cybersecurity](#) (completed cohort 1 with 23 participants)
- Co-Organiser, [2024 JTG/IEEE ITSoc Summer School](#) (24–28 JUN 2024)

PROFESSIONAL SERVICE

- Reviewer, Transactions on Machine Learning Research (TMLR), 2026
- Reviewer, IEEE International Conference on Computer Communications (Infocom 2026)
- Technical Program Committee (TPC) Member, ISIT 2025
- Reviewer, IEEE Transactions on Information Theory
- Reviewer, IEEE Transactions on Signal Processing
- Reviewer, Journal on Selected Areas in Communications (JSAC)
- Reviewer, International Conference on Machine Learning (ICML 2024)
- Reviewer, International Conference on Learning Representations (ICLR 2024)
- Reviewer, Conference on Neural Information Processing Systems (NeurIPS 2023, 2024, 2026)
- Reviewer, Conference on Decision and Control (CDC 2023)
- Reviewer, IEEE International Symposium on Information Theory (ISIT 2019, 2023, 2024)
- Reviewer, National Conference on Communications (NCC 2018)